

OS Credential Dumping: LSASS Memory (T1003.001)

August 17, 2026

1 Attack Description in CNL

The attack description of the MITRE ATT&CK Technique “*OS Credential Dumping: LSASS Memory*” (T1003.001)¹ written in the proposed CNL is shown below.

```
1 Tactic: TA0006 Credential_Access
2 Technique: T1003.001 OS_Credential_Dumping_LSASS_Memory
3
4 Background
5 Given process as abnormal_process
6 And process as lsass_exe with command_line = "lsass.exe", signed = "true"
7 And handle as lsass_handle with hexadecimal_number = "0x1F0FFF", target = "lsass.exe"
8 And file as lsass_dump
9 And registry as registry_keys
10
11 Event 1
12 Given abnormal_process is activated And lsass_exe is activated
13 When abnormal_process requests lsass_handle
14 Then lsass_handle acquired by abnormal_process
15
16 Event 2
17 Given lsass_handle acquired by abnormal_process
18 When abnormal_process requests lsass_dump
19 Then lsass_dump is created
20
21 Event 3
22 Given lsass_dump is created
23 When abnormal_process modifies registry_keys
24 Then registry_keys modified
25
26 Completion
27 Event 2 Xor Event 3
```

2 Attack Model (Petri Net)

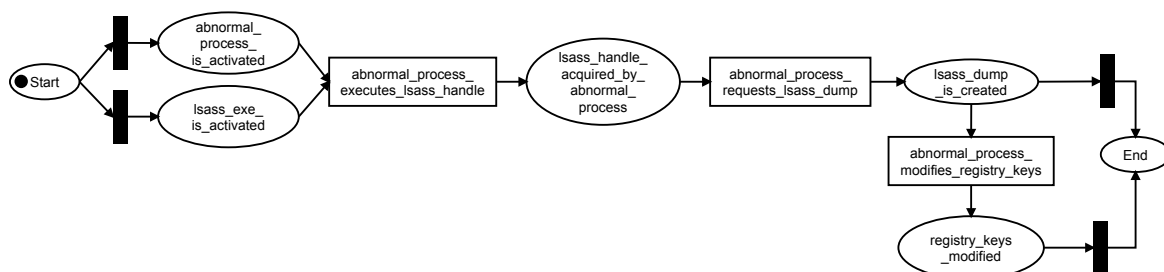


Figure 1: OS Credential Dumping: LSASS Memory Petri Net

¹See <https://attack.mitre.org/techniques/T1003/001/>

3 Runtime and Memory Usage

The translation of this CNL description took 464ms and allocated 554KB at peak. The translation pipeline was run on a laptop with Intel Core i7-12650H, 16 GB RAM and running Windows 11.

4 Validation Criteria

The validation process included semi-structured interviews with two domain experts, in which the following validation criteria were discussed: **V1.** the technique description (in CNL) expresses the content of the ATT&CK text; **V2.** the technique description (in CNL) does not contain ambiguities; **V3.** the Technique Model captures the described attack features; **V4.** the Petri Net presents the alternative paths the attack can follow based on the described actions, pre- and post-conditions and assets involved. In the discussion section below, the domain experts are referred to as E1 and E2.

5 Discussion

5.1 V1 & V2

The attack description of the technique fully expresses all attack features present in the source description. The expressions are clear and match the high level of abstraction the source description is written in, while reducing ambiguities that hinder the interpretation of the description in terms of the technique’s execution. Assets are declared as they appear in ATT&CK, with their attributes stated explicitly. The handle to *lsass.exe* for example, is declared as an asset in the description, carrying the hexadecimal number (access mask) `0x1F0FFF` and a `target` reference to the *lsass.exe* process. The connection between the handle and its mask `0x1F0FFF` is briefly stated in the source description, but the encoding makes it a explicit reference that the events can refer to. During the interview, E1 noted that assets are still abstract in the CNL description, causing their declaration to be less specific. The actual identity of the assets in a real system is not present in the CNL description by design, since the source description also does not explicitly state it. For example “*abnormal process*” is not further explained. Its potential integration into a detection system would require further enrichment of the asset. However, for the scope of this work, the CNL description matches the content of the source description and does not introduce new ambiguities. E1 also highlighted that the proposed CNL allows the attack to be described in a way that is operational, without compromising human readability: “*It (the CNL) maintains the natural language, while being operational*”.

5.2 V3 & V4

The Technique Model successfully captures the attack features exactly as stated in the CNL description in their appropriate types and composition. The Petri Net presents the system behavior under the attack, showing the interconnection between the actions and the alternative paths the attack can follow. Additionally, the names of the transitions and places match the verbs and the names of the involved assets as stated in the events and state conditions. During the second interview, E2 stated that “*The language forces me to be specific about the actions of the attack*” and “*here (in the Petri Net) I immediately see how the technique plays out*”. They also observed that the proposed CNL could potentially assist in the identification of vulnerable assets: “*I can directly see which are the vulnerable assets the attack targets*”.